CODECHEF

The CodeChef.com Challenge—your monthly dose of coding puzzles from India's biggest coding contest, now in print!

Hello and welcome to the CodeChef Puzzle Challenge sponsored by LINUX For You and Directi! Let's see how many of you can solve this month's puzzle, and find out which one of you will be amongst the three lucky winners of a cash prize of Rs 1000!

August edition's puzzle

Ten friends start from their respective homes at the same time to meet any one of the other friends. They choose to meet anyone with equal likelihood. All of them take equal time to reach the other friend's home, and given that none of them use the same route, what is the probability that none of them meet.

The solution

The solution for N=10 is 0.00038286307568

How did we do it?

The solution is based on permutations with inclusion-exclusion principle. Accordingly it will be:

(N! * (1/0! - 1/1! + 1/2! - 1/3! +... (-1) ^ N/N!)) / (N-1) ^ N Refer: http://en.wikipedia.org/wiki/Derangement

And the winners for August are...

- Arjun Pakrashi
- Gowtham Varma
- Bharadwaj Varma

Your Codechef Puzzle Of The Month

Sometime back, Donald Knuth (if you don't know him then you need not read any further) made an "Earthshaking Announcement" at the TeX 32nd Anniversary Celebration. Contrary to people expecting he'd solved the P!=NP problem, it turned out to be a joke. Apu, a Comp. Sci. student from India with no sense of humour, did not take this lightly and decided to solve the problem himself. Then he realized Vinay Deolikar had already solved the problem. So, Apu decided to solve the 'Sorting Problem' in less than O(N * log(N)) time using something called a k-ordered array.

An array A[1...n] is said to be k-ordered if A[i - k] leq A[i] leq A[i+ k] for all i such that

k leq i < n - k

k < i leq n - k.

However, Apu is now stuck. What is the maximum possible distance between two numbers which are out of sort, given a very long array an array which is both 5-ordered and 7-ordered? (Distance implies the difference of indexes of the numbers. Eg. in this 2-ordered array [4, 1, 5, 2, 6, 3] the maximum distance is 5) Can you help him out?

Looking For A Little Extra Cash?

Solve this month's CodeChef Challenge puzzle, send in your answer to codechef@efyindia.com and you may just be one of our three lucky winners to wins Rs.1000 in cash!

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About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to www.codechef.com for the October Challenge that takes place from October 1st to the 11th, 2010 to win cash prizes up to Rs.20,000. To find out how the Directi 'Go For Gold' initiative aimes to help India win the ACM ICPC (international collegiate programming contest) World finals, visit: www.codechef.com/goforgold